PHY102 Electricity and Magnetism Jan-April 2026: Assignment 5

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To be discussed in the Tutorial on March 3

1. Consider thin glass pipe of length L and cross section area A laying along the x axis with one end at origin and the other end at $x = L$. There is vacuum inside the pipe. Electrons enter the end at $x = 0$ at almost zero velocity (You can think about them coming from the thermionic emission). The voltage V is applied across the pipe to accelerate and collect the electrons on the other end at $x = L$. The circuit is completed and the steady state current is I(Same current I folows through the wire and through the pipe). If λ defines the linear charge density i.e. charge per unit length inside the pipe. Find λ as a function of x and plot its graph. How do you understand this graph.

2. Consider a parallel plate capacitor with plate area A and distance d between the plates. Calculate the energy stored in the capacitor when the capacitor is charged to a voltage V i.e. the voltage difference between the plates is V.

3. Imagine the image charge problem discussed in the class where a positive charge Q is placed at a distance h above a conducting plane. What is the work done in moving this charge from its current location h to infinite distance from the plane. Do this calculation with and without using the image charge.

4. Consider an infinite line charge with linear charge density λ laying along x axis at rest in Frame F_0. How would this line charge density appear from a frame F_1 moving with a velocity v along positive x direction and from a frame F_2 moving with velocity v along the positive y direction. Now consider a an infinite charge sheet in the $y - z$ plane with surface charge density σ at rest in F_0. How does this charge density looks from the frames F_1 and F_2. It is useful to use dimensionless parameters $\beta = v/c$ and $\gamma = \frac{1}{\sqrt{1-\beta^2}}$